

Constraint-flag breakdown for finite-support optimal-Wannier resets: a preregistered CPU pilot

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We formulate a finite-size diagnostic for the obstruction encountered when an adaptive Gaussian circuit attempts to impose a redundant word of compactly truncated optimal-Wannier (OW) occupation constraints. Filled and empty reset modes generate two increasing subspace flags and hence a decreasing filtration of compatible half-filled Grassmannians. Exact failure is detected by a rank bound or a nonzero principal angle overlap; approximate failure is quantified by a Ky Fan minimum-frustration functional. We sharply separate this static completion problem from a cumulative tangent-kernel flag and from the dynamical loss of a reset constraint upon its next visit. A preregistered 16×20 CPU pilot compares a support-terminated domain-wall slab to a periodic uniform control under random and raster schedules. The construction evades no known no-go theorem: finite support removes the exponentially localized common dark state required for exact pure Chern preparation, while the conditional trajectory ensemble and unconditional mixed state are logically distinct objects.

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I. PREMISE AND RELATION TO THE DISSIPATIVE NO-GO THEOREM

An exactly flat Chern band admits exponentially localized OW modes but no translation- covariant compactly supported orthonormal Wannier basis. Goldstein’s dissipative no-go theorem makes the corresponding preparation tension operational: a finite-range, number-conserving quadratic Lindbladian cannot simultaneously have a unique pure Chern steady state and a size-independent dissipative gap under its stated regularity assumptions [1]. Our circuit uses finite n_{shell} truncations of otherwise exponential OW tails. Those compact controller modes need not share any exact half-filled Gaussian dark state. Perfect correction resets the mode selected at the current letter; it does not make all earlier nonorthogonal occupation constraints simultaneous constants of motion.

There is therefore no evasion of the obstruction. At finite support, one relinquishes the common exact pure Chern fixed point. Along a conditioned trajectory the state remains Gaussian and pure after the measurement/reset record is specified, whereas the unconditional state is a record average and is generally mixed. In a hard-wall slab, the slowest restoration scale may also close with system size. The present experiment asks where this incompatibility first becomes visible and whether its excess is localized at the support-termination boundary. It does *not* use overlap winding as the compact mode’s Chern number.

II. FILLED/EMPTY FLAGS AND COMPATIBLE COMPLETIONS

Let $\mathcal{H} \simeq \mathbb{C}^D$ be the active one-body space and let a circuit word be $(w_j, s_j)_{j=1}^M$, where $\|w_j\| = 1$ and $s_j \in \{0, 1\}$ is the target occupation. Define the increasing flags

$$\mathcal{F}_m = \text{span}\{w_j : s_j = 1, j \leq m\}, \quad \mathcal{E}_m = \text{span}\{w_j : s_j = 0, j \leq m\}. \quad (1)$$

A pure number-conserving Gaussian state with N particles is an occupied plane $U \in \text{Gr}(N, D)$, or equivalently a rank- N orthogonal projector P_U . Its compatible set after the first m letters is

$$\mathfrak{M}_m = \{U \in \text{Gr}(N, D) : \mathcal{F}_m \subseteq U \subseteq \mathcal{E}_m^\perp\}. \quad (2)$$

Thus $\mathfrak{M}_{m+1} \subseteq \mathfrak{M}_m$: the word generates a decreasing Grassmannian filtration even though $\mathcal{F}_m$ and $\mathcal{E}_m$ increase.

Proposition 1 (Exact completion). *The prefix admits a rank- N completion if and only if*

$$\mathcal{F}_m \perp \mathcal{E}_m, \quad \dim \mathcal{F}_m \leq N \leq D - \dim \mathcal{E}_m. \quad (3)$$

Proof. Necessity follows because $\mathcal{F}_m \subset U$ and $\mathcal{E}_m \subset U^\perp$. Conversely, orthogonality gives $\mathcal{F}_m \subset \mathcal{E}_m^\perp$. The rank bounds ensure that one may choose an $(N - \dim \mathcal{F}_m)$ -plane in $\mathcal{E}_m^\perp \ominus \mathcal{F}_m$ and append it to $\mathcal{F}_m$. $\square$

The exact breakdown letter is

$$m_\star = \min\{m : \mathfrak{M}_m = \emptyset\}. \quad (4)$$

With orthonormal span matrices $Q_{F,m}$ and $Q_{E,m}$, nonorthogonality is diagnosed by $\sigma_{\max}(Q_{F,m}^\dagger Q_{E,m}) > 0$. We also record the total principal-angle overlap

$$\Phi_m = \|Q_{F,m}^\dagger Q_{E,m}\|_F^2 = \sum_a \cos^2 \theta_a. \quad (5)$$

III. MINIMUM FRUSTRATION AND MARGINAL COST

Exact feasibility is numerically brittle and deliberately binary. For normalized constraints and nonnegative weights γ_j , define

$$F_m(P) = \sum_{j \leq m} \gamma_j [s_j(1 - \langle w_j, P w_j \rangle) + (1 - s_j) \langle w_j, P w_j \rangle], \quad P^2 = P = P^\dagger, \quad \text{tr } P = N. \quad (6)$$

Writing $A_m = \sum_{j \leq m} \gamma_j (1 - 2s_j) |w_j\rangle \langle w_j|$, the Ky Fan principle gives

$$F_\star(m) = \sum_{j \leq m} \gamma_j s_j + \sum_{a=1}^N \lambda_a^\uparrow(A_m), \quad \Delta_m = F_\star(m) - F_\star(m-1) \geq 0. \quad (7)$$

The implementation updates numerical span bases and rank conditions at every letter. It evaluates principal spectra and Eq. (7) at preregistered block, shell, column, and random-word decile checkpoints; at each such m , $F_\star(m-1)$ is also evaluated so the reported Δ_m is the literal one-letter marginal.

IV. THREE OBJECTS THAT MUST NOT BE CONFLATED

The *static constraint flag* above asks for a simultaneous Gaussian completion of a finite word. A distinct *cumulative tangent-kernel flag* is formed from kernels of linearized measurement/reset maps acting on covariance perturbations. Its rank loss measures erased tangent directions and can be studied later, but is not inferred here. Finally, *dynamical reset retention* is a revisit statistic. Immediately before the next imposition of letter j in cycle t , let

$$\epsilon_{j,t} = 1 - p_{\text{target}}(j, t). \quad (8)$$

The observed wrong outcome $X_{j,t} \in \{0, 1\}$ obeys $\mathbb{E}[X_{j,t} \mid \mathcal{F}_{t-}] = \epsilon_{j,t}$. Therefore $1 - \epsilon$ is retention, and $\langle X \rangle = \langle \epsilon \rangle$ is a mandatory calibration check rather than an additional physical hypothesis. Static incompatibility does not by itself imply persistent stationary activity.

V. PREREGISTERED CPU PROTOCOL

The production pilot uses $N_x \times N_y = 16 \times 20$, $n_{\text{shell}} = 1$, perfect correction, and the maximally mixed initial covariance. The hard support-terminated wall has active centers $x = 3, \dots, 13$; the static uniform comparison uses the same centers but retains the full periodic one-body basis. Static $n_{\text{shell}} = 2$ and untruncated frames are convergence controls. We order constraints from interior shells to the walls, in raster- y order, and by each saved first-cycle random word.

Dynamics compare the wall and uniform geometries under random and raster- y schedules. Ten root seed IDs are reused across configurations and are called *matched-seed* IDs, not identical records, because active words differ. Each trajectory has 80 cycles; cycles 1–40 are burn-in. The statistical unit is the trajectory, never the individual site/channel revisit. We report paired matched-seed and unpaired trajectory-bootstrap intervals, left/right wall regions, the interior, channel and position in the word, and uniform pseudo-interfaces at $x = 3, 13$.

VI. PROMOTION RULE AND LIMITATIONS

GPU/Colab work is recommended only if all of the following hold: (i) a reproducible static marginal-frustration excess begins when boundary shells enter; (ii) it is present at both walls and above the uniform finite-shell background; (iii) post-burn-in revisit loss is greater at both wall interfaces than in the wall interior for both schedules; (iv) uniform pseudo-interfaces have no comparable peak; and (v) signs survive both matched-seed and unpaired trajectory bootstrap analyses. Otherwise the pilot is reported as null or schedule-dependent and no GPU campaign is started.

A flag defect is not by itself proof of topology, chirality, a compact-mode Chern number, or stationary activity. Finite size, the chosen OW gauge, support termination, and schedule all remain possible sources of structure. The experiment resolves a specific compatibility mechanism, not an invariant classification.

VII. GENERATED PILOT RESULT

Pilot pending. The analysis stage will replace this fragment atomically with the measured scalar table, gate verdict, and provenance-linked figures.

Status: pilot pending. Parameters: 16×20 , 10 matched seed IDs, 80 cycles, burn-in 40. Gate verdict: `PILOT_PENDING`.

[1] G. Goldstein, “Dissipation-induced topological insulators: A no-go theorem and a recipe,” SciPost Phys. **7**, 067 (2019).